HX-500 CNC Machining Centre: Maintenance Manual (Rev. 4)

This manual covers routine care of the HX-500 vertical machining centres in Bay 2. Operators do the daily checks; the maintenance team does the weekly and monthly work.

Daily checks, start of every shift: check the way-lube reservoir is above the MIN line, drain water from the air filter bowl, confirm air pressure reads between 0.5 and 0.7 MPa, and wipe chips off the tool-changer arm.

Spindle warm-up: before the first job of the day, or after the machine has been idle for more than 4 hours, run warm-up program O9001. It runs the spindle for 10 minutes at 30% of maximum speed, then 5 minutes at 60%. Skipping warm-up shortens bearing life and causes size drift on the first parts.

Coolant

Coolant is a water-soluble semi-synthetic. Concentration must be kept between 6% and 8%, measured with the refractometer at the start of each day shift and written in the coolant log.

Below 6% the risk of rust and bacterial growth rises; above 8% causes foaming and skin irritation. Top up with premix from the central mixer, never with neat concentrate or plain water.

The HX-500 coolant tank holds 380 litres. Clean the tank and replace the coolant every 6 months, or sooner if it smells sour.

Alarm codes

AL-214 Spindle overheat: stop the program, leave the spindle off for at least 20 minutes, and check the chiller is running and its filter is clean. If the alarm returns within one shift, call maintenance.

AL-305 Tool length mismatch: the measured tool length differs from the offset table by more than 0.05 mm. Re-measure the tool on the presetter; do not override.

AL-410 Low air pressure: pressure below 0.5 MPa. Check the main supply valve and the filter bowl.

AL-522 Door interlock open: the guard door was opened during a cycle. Close it and press Reset; never bypass the interlock.